

Assignment 8: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

NOTE: For this assignment, you can skip tests for any of the assumptions required to run the models.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
#set up session
#load in packages
library(tidyverse); library(here); library(agricolae)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## here() starts at /home/guest/ENV872/EDE_Spring2026/Assignments
```

```
library(lubridate); library(dplyr); library(ggplot2)
#dplyr to use complete.cases command later
here() #check working directory
```

```
## [1] "/home/guest/ENV872/EDE_Spring2026/Assignments"
```

```
#returns "/home/guest/ENV872/EDE_Spring2026/Assignments"
#import raw data file
raw_data <- read.csv(
  here("Data/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"))
class(raw_data$sampldate)
```

```
## [1] "character"
```

```
#set date columns to date objects
raw_data$sampldate <- as.Date(raw_data$sampldate, format = "%m/%d/%y")
#check that conversion to date object was successful
class(raw_data$sampldate)
```

```
## [1] "Date"
```

```
#2
#build a ggplot theme and set as default
mytheme <- theme_classic(base_size = 14) +
  theme(
    axis.text = element_text(color = "black"),
    axis.title = element_text(size = 12),
    axis.line = element_line(linewidth = 0.8),
    legend.position = "top",
    legend.title = element_text(face = "bold"),
    legend.key = element_blank(),
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
  )
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperature recorded during July does not change with depth across all lakes (slope of depth vs mean July lake temperature is zero). Ha: Mean lake temperature recorded during July changes with depth across all lakes (slope of depth vs mean July lake temperature is nonzero).
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

- Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

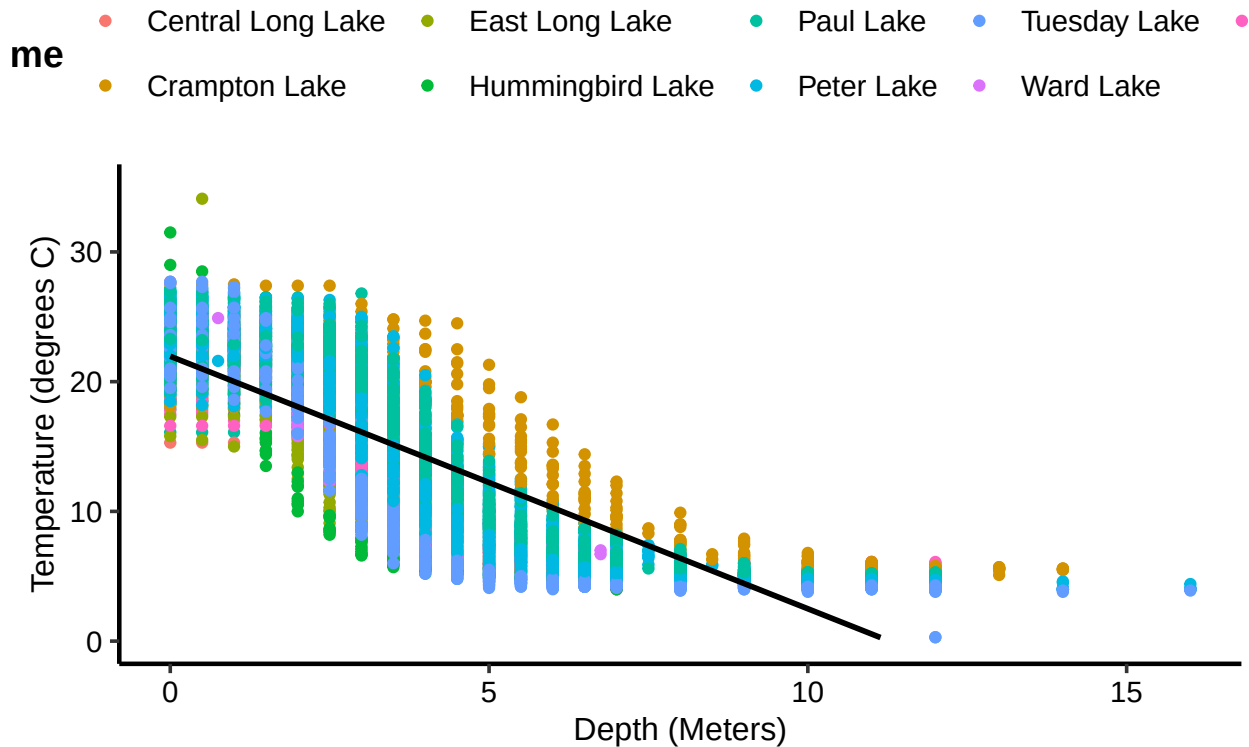
```
#4
#wrap the dataset
NTL_LTER_wrangled <-
  raw_data %>%
  mutate(Month = month(sampledate)) %>% #parse date column, add month column
  filter(Month == 7) %>% #keep only dates in July
  select("lakename", "year4", "daynum", "depth", "temperature_C") %>%
  #keep only certain columns
  filter(complete.cases())

#5
#create scatter plot of temp by depth
temp_by_depth_plot <- NTL_LTER_wrangled %>%
  ggplot(aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point() +
  geom_smooth(aes(group = 1), method = "lm", color = "black") +
  #add only one smoothed line showing linear model as per instructions
  scale_y_continuous(limits = c(0,35)) + #limit temp values from 0 to 35
  labs(
    title = "Temperature by Depth of Lake",
    x = "Depth (Meters)",
    y = "Temperature (degrees C)",
    color = "Lake Name")
#add labels, and color code points by lake
print(temp_by_depth_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Temperature by Depth of Lake



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The distribution of points suggest a linear relationship between the two variables. It appears that temperature decreases with increasing depth. This trend is strongest at a depth of between 0-10 meters.

7. Perform a linear regression to test the relationship and display the results.

```
#7
#perform a linear regression
regression <-
  lm(NTL_LTER_wrangled$temperature_C ~ NTL_LTER_wrangled$depth)
print(regression) #displays the coefficients

##
## Call:
## lm(formula = NTL_LTER_wrangled$temperature_C ~ NTL_LTER_wrangled$depth)
##
## Coefficients:
##              (Intercept)  NTL_LTER_wrangled$depth
##                21.956             -1.946
```

```

#show the *strength* of the relationship
cor.test(NTL_LTER_wrangled$temperature_C, NTL_LTER_wrangled$depth)

##
## Pearson's product-moment correlation
##
## data: NTL_LTER_wrangled$temperature_C and NTL_LTER_wrangled$depth
## t = -165.83, df = 9726, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.8646036 -0.8542169
## sample estimates:
## cor
## -0.8594989

#prints a p value of < 0.05, indicating that the null hypothesis is not supported
#cor is -0.859, strong negative relationship
summary(regression)

##
## Call:
## lm(formula = NTL_LTER_wrangled$temperature_C ~ NTL_LTER_wrangled$depth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    21.95597    0.06792   323.3  <2e-16 ***
## NTL_LTER_wrangled$depth -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The cor value given by the Pearson's product-moment correlation is -0.859, therefore the r squared value is 0.738. Therefore, 73.8% of variation in temperature can be explained by changes in depth. This finding is based on 9726 degrees of freedom (df=9726), and the p-value is 2.2e-16, indicating that the relationship between temperature and depth is statistically significant and we should reject the null hypothesis that mean lake temperature does not change with depth. Furthermore, the slope is -1.946. This means that for every 1m change in depth, the temperature is expected to decrease by 1.946 degrees Celsius.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
#run AIC:
#determine which set of explanatory variables is best suited to predict temp

# create model object with predictors
TPAIC <- lm(temperature_C ~ lakenam + year4 + daynum + depth,
            data = NTL_LTER_wrangled)

# Stepwise model selection using AIC
TPAIC_step <- step(TPAIC)
```

```
## Start: AIC=24516.31
## temperature_C ~ lakenam + year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 120629 24516
## - year4      1         168 120797 24528
## - daynum     1        1296 121925 24618
## - lakenam    8        21058 141687 26066
## - depth     1       403300 523930 38801
```

```
# View selected model
summary(TPAIC_step)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakenam + year4 + daynum + depth,
##     data = NTL_LTER_wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.892  -3.036  -0.218   2.779  15.284
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    43.837042   8.246592   5.316 1.09e-07 ***
## lakenamCrampton Lake    4.762283   0.373374  12.755 < 2e-16 ***
## lakenamEast Long Lake  -1.403887   0.333137  -4.214 2.53e-05 ***
## lakenamHummingbird Lake -4.676827   0.452809 -10.328 < 2e-16 ***
## lakenamPaul Lake       1.062414   0.321114   3.309 0.000941 ***
## lakenamPeter Lake      1.500793   0.320895   4.677 2.95e-06 ***
## lakenamTuesday Lake   -1.314121   0.325965  -4.031 5.59e-05 ***
```

```
## lakenameward Lake      -0.418098    0.457643   -0.914 0.360955
## lakenamewest Long Lake -0.124557    0.332107   -0.375 0.707631
## year4                  -0.015165    0.004119   -3.682 0.000233 ***
## daynum                  0.040749    0.003989   10.217 < 2e-16 ***
## depth                  -1.964288    0.010899  -180.232 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.524 on 9716 degrees of freedom
## Multiple R-squared:  0.7797, Adjusted R-squared:  0.7794
## F-statistic: 3125 on 11 and 9716 DF, p-value: < 2.2e-16
```

```
#The model retained year4, daynum, and depth, indicating that all of these
#variables are suited to predict temperature.
#The p value for all three of these variables is less than 0.05, as shown in the
#4th column.
```

```
#10
#run a multiple regression on the recommended set of variables
multiple_regression <- lm(data = NTL_LTER_wrangled,
                          temperature_C ~ year4 + daynum + depth)
summary(multiple_regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER_wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

```
#summary outputs a multiple R-squared value of 0.7412
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression is lakenameward, year4, daynum, and depth. The AIC output

shows that removing any of these increases AIC. Please note that in my multiple regression, I only included year4, daynum, and depth since this is how I interpreted the instructions in Question 9. Since our multiple r-squared value is 0.7797, we can conclude that 77.97% of the variation in temperature is explained by the model consisting of these four variables. This is a minor improvement over the model using only depth, because the r squared for that model was 0.738.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
#create two sets of models to see whether lakes have different temps in July

# Format ANOVA as aov
ANOVA_aov <- aov(data = NTL_LTER_wrangled, temperature_C ~ lakename)
summary(ANOVA_aov)

##               Df Sum Sq Mean Sq F value Pr(>F)
## lakename       8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#results: p value below 0.05 so reject null hypothesis i.e. difference between a
#pair of group means is statistically significant

# Format ANOVA as lm
ANOVA_lm <- lm(data = NTL_LTER_wrangled, temperature_C ~ lakename)
summary(ANOVA_lm)

##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTL_LTER_wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake     -6.5972     0.6769  -9.746 < 2e-16 ***
```

```
## lakenameward Lake      -3.2078      0.9429  -3.402 0.000672 ***
## lakenamewest Long Lake -6.0878      0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

```
#results: p value below 0.05 so reject null hypothesis i.e. there are
#significant differences in temp between lakes
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: Yes, due to the p values being below 0.05 in both models, we can conclude that there is a significant difference in mean temperature among the lakes.

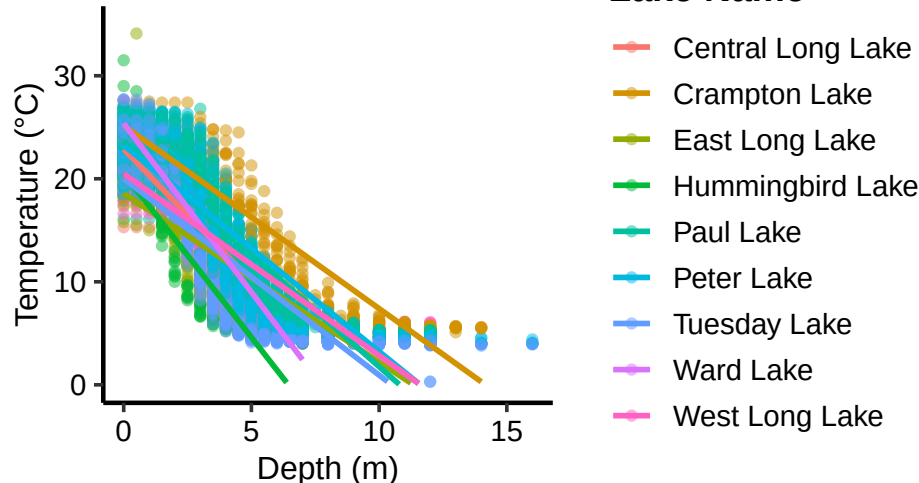
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
#create a graph that depicts temperature by depth
temperature_by_depth_plot <- ggplot(NTL_LTER_wrangled, aes(x = depth,
  y = temperature_C, color = lakenamewest)) +
  geom_point(alpha = 0.5) + # 50% transparent points
  geom_smooth(method = "lm", se = FALSE) + # regression line for each lake
  scale_y_continuous(limits = c(0, 35)) + # y-axis limits
  labs(
    title = "Temperature vs Depth by Lake",
    x = "Depth (m)",
    y = "Temperature (°C)",
    color = "Lake Name") +
  theme(legend.position = "right")
show(temperature_by_depth_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Temperature vs Depth by Lake



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
#run Tukey's HSD test
TukeyHSD(ANOVA_aov)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = NTL_LTER_wrangled)
##
## $lakename
##
##              diff              lwr              upr      p adj
## Crampton Lake-Central Long Lake -2.3145195 -4.7031913  0.0741524 0.0661566
## East Long Lake-Central Long Lake -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake      -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake      -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake    -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake       -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake  -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake      -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake    -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake           -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake          -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake        -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake           -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake      -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake   0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake          3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake         3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake        0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake          4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake     1.3109897  0.2885003  2.3334791 0.0022805
```

## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

*#the results indicate that many lakes have different means, since the p adj is
#below 0.05 for many of the comparisons.*

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: The lakes that have the same mean temperature, statistically speaking, as Peter Lake are Paul Lake and Ward Lake. Peter Lake-Paul Lake has a p-value of 0.224 while Peter Lake-Ward Lake has a p value of 0.782. No single lake has a mean temperature that is statistically distinct from all the other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: Looking at Peter Lake and Paul Lake, we could also use a two sample t test to see whether they have distinct mean temperatures. A two sample t test uses the hypothesis that the mean of two samples is equivalent.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
#wrangle July data to include only records for Crampton Lake and Ward Lake
NTL_LTER_wrangled_2 <-
  NTL_LTER_wrangled %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

Ttest <- t.test(temperature_C ~ lakename, data = NTL_LTER_wrangled_2)
print(Ttest)
```

```
##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
```

```
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to zero
## 95 percent confidence interval:
##  -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189                14.45862
```

Answer: The test states that the mean in Crampton lake is 15.35 and the mean in War Lake is 14.46, and provides a 95 percent confidence interval for the difference in means. Most importantly, the p-value is $0.2649 > 0.05$, meaning we cannot reject the null hypothesis. The p-value fails to reject our null hypothesis that the true difference in means between Crampton lake and Ward lake is equal to zero. This is consistent with our answer to question 16 that no single lake has a mean temperature that is statistically distinct from all the other lakes.